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Established machine learning matches tabular foundation models in clinical predictions

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Abstract

Foundation models (FMs) promise to standardise predictive modeling across domains, yet their clinical value for tabular data remains unproven. To test this, we performed a large, fully reproducible benchmark of TabPFN, a leading FM for tabular prediction, against twelve established machine learning (ML) methods across twelve binary clinical tasks. Cohorts spanned 788 – 139,528 patients across diverse outcomes, including survival, metastasis, and disease status. Using standardized preprocessing, bootstrapping, and multiple performance metrics, TabPFN was generally competitive but did not consistently outperform strong ML baselines. It exceeded the best ML model in only 16.7% of tasks, with most area under the receiver operating characteristic (AUROC) differences within ±0.02. TabPFN also incurred higher computational cost, with median runtimes 5.5× longer and practical reliance on GPU acceleration. These findings indicate that, for routine binary clinical prediction, TabPFN offers limited performance gains relative to optimized ML methods, while introducing significant efficiency trade-offs.

Keywords Tabular data, Machine learning, Deep learning, TabPFN, Benchmarking

Introduction

Artificial intelligence (AI), including its applications in data science, is rapidly evolving, and recent advances have led to the development of large-scale, pre-trained models known as foundation models (FMs). FMs are trained through a self-supervised-learning method [1], in which the model takes large amounts of unlabeled data and repeatedly tries to learn meaningful patterns that can then be adapted to a wide range of downstream tasks such as classification or prediction [2]. In clinical applications, such models have already been applied to detect meaningful patterns from complex datasets, particularly in domains such as pathology [3], radiological imaging [4], and clinical text [5]. The inclusion of FMs in digital pathology workflows has accelerated research [6] and is beginning to extend into clinical practice. Unlike conventional machine learning (ML) models, which often require feature engineering and large, task-specific labeled datasets to be developed from the ground up, FMs are pre-trained on massive, heterogeneous datasets and can often be fine-tuned with fewer annotations while achieving performance comparable to, or exceeding, that of task-specific ML models [7].

Like medical images and clinical text, electronic health record (EHR) data are widely used in healthcare and often capture essential information related to patient histories, diagnoses, treatments, and outcomes. A substantial proportion of clinical data is structured or semi-structured

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in tabular form, with defined variables such as demographics, laboratory results, medications, and diagnostic codes routinely recorded in rows and columns. An ML model or FM trained for this data could therefore offer similar value in clinical settings, as well as new insights, by identifying patterns that may not be readily apparent using conventional approaches.

A leading foundation model for tabular prediction is the Tabular Prior-data Fitted Network (TabPFN), a generative, transformer-based FM introduced and developed by Prior Labs (Prior Labs GmbH, Germany). Hollmann et al. reported that TabPFN outperformed state-of-the-art baselines for tabular prediction including traditional ML models such as XGBoost, as well as a range of supervised deep learning architectures tailored for tabular prediction such as TabNet, DNF-Net, and FT-Transformer [8]. Unlike conventional ML models that require retraining for each new task, TabPFN performs in-context learning, enabling single-pass predictions without parameter updates. It processes training and testing data using a two-way attention mechanism, allowing it to focus on relevant patterns at inference time by capturing relationships across columns. Currently TabPFN is optimized for predictions in small to medium sized datasets, although future versions promise to exceed these limitations (https://priorlabs.ai/tabpfn/large-data). TabPFN was trained on millions of synthetic datasets from varied domains and benchmarked against widely used prediction models; Hollmann et al. reported improvements of approximately 0.13–0.19 in per-dataset normalized area under the receiver operating characteristic (AUROC) when compared to the strongest baselines [8]. However, although TabPFN was pre-trained on synthetic datasets and evaluated on general tabular benchmarks, its ability to generalize across heterogeneous clinical tasks and outperform established ML models remains untested. If performance gains are small or inconsistent, the added complexity and computational costs may not justify replacing established tabular ML pipelines.

Previous studies comparing TabPFN with various ML models have reported mixed results. Karabacak et al. conducted twelve independent benchmarking studies on clinical datasets, in which ML models outperformed TabPFN in five cases [9–13], while TabPFN performed best in three [14–16], and performance was task dependent in the remaining four [17–20]. At the time of writing, nine additional published studies similarly report heterogeneous outcomes, with four favoring TabPFN [21–24], two favoring ML [25, 26], and three suggesting context-specific performance [27–29]. However, many of these studies rely on private datasets and heterogeneous evaluation pipelines, including differences in feature handling, missing-data strategies, imputation, encoding, class imbalance assessment, resampling, and model tuning. As a result, reported performance differences may reflect pipeline choices as much as underlying model capability.

In this study, we compare established ML models to the TabPFN FM in terms of performance and efficiency to evaluate the feasibility of replacing established ML models with this FM. TabPFN was benchmarked against strong representatives of the major model families commonly used for clinical tabular prediction, including classic statistical models [30], bagging ensembles [31], boosting methods [32], and multilayer perceptrons [33]. These families reflect distinct trade-offs in robustness for small-sample tabular data, calibration, interpretability, and computational cost, and together provide a more informative baseline than any single model class. In addition, we quantify runtime on defined hardware and evaluate TabPFN with its recently introduced hyperparameter optimization (HPO) feature. By applying the same pipeline to every target, we obtain performance and efficiency metrics that are directly comparable across models and datasets. This rigorous, uniform framework was essential to isolate genuine differences in model capability from artifacts introduced by heterogeneous analysis pipelines.

Results

We benchmarked TabPFN against twelve established ML models across twelve binary clinical prediction tasks from eight publicly available datasets, assessing discrimination performance, reproducibility relative to published results, and computational efficiency.

Benchmark scope and task heterogeneity

The benchmark comprises twelve clinically relevant binary classification tasks derived from three independent data sources: the Fundação Oncocentro de São Paulo (FOSP) registry, public datasets curated by Arasteh et al. [34], and the Surveillance, Epidemiology, and End Results (SEER) cancer registry (Fig. 1a, b). All tasks are based on structured patient-level tabular data using consistent variable representations across datasets (Fig. 1a).

Across tasks, cohort sizes ranged from 788 to 139,528 patients, feature dimensionality varied from 15 to 1,877 variables, and target prevalences spanned a wide range from rare outcomes (2.5%; esophageal cancer) to common events (72.0%; 1-year survival) (Fig. 1b). The benchmark includes both cancer-related and non-cancer prediction tasks, as well as disease status classification and fixed-horizon survival endpoints.

All datasets were evaluated using a unified benchmarking workflow, including standardized preprocessing, model training across multiple ML families, and repeated bootstrap evaluation with consistent performance metrics (Fig. 1c). This design enables direct comparison of model performance across heterogeneous clinical

a.

FOSP ···

Date of Diagnosis	ICD-O-3	Morphology	Sex	1 Year Survival
2000-01-04	C019	80703	M	Yes
2000-10-02	C219	80703	M	Yes
2000-12-18	C100	80703	F	No
2002-05-25	C446	87203	M	Yes

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b.

Cancer-target	FOSP	Patients	Features	Positive targets
Yes	1 Year Survival	91,795	72	72.0%
Yes	3 Year Survival	91,795	72	27.2%
Yes	5 Year Survival	91,795	72	20.4%
Yes	Cancer Death	91,795	72	39.7%
No	Overall Death	91,795	72	55.9%
Cancer-target	**Arasteh et al.**	**Patients**	**Features**	**Positive targets**
No	Amyloidosis	2,142	1,877	50.0%
Yes	Esophageal Cancer	14,597	121	2.5%
No	Hereditary Hearing Loss	1,778	144	24.1%
Yes	Metastatis Disease	788	15	28.3%
Cancer-target	**SEER**	**Patients**	**Features**	**Positive targets**
Yes	HCC to Lung Met	62,869	82	6.5%
Yes	Osteosarcoma	4,097	73	33.8%
Yes	Renal Cell Carcinoma	139,528	78	34.3%

c.

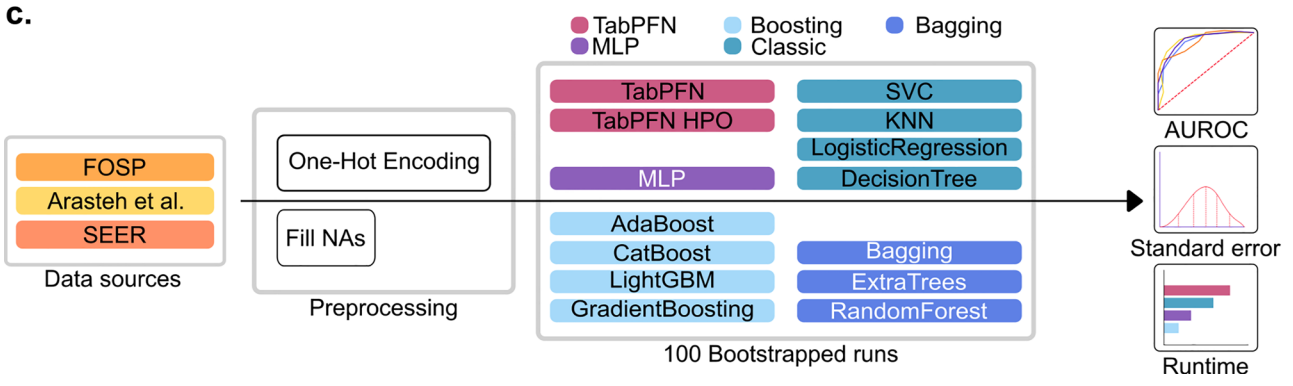


Fig. 1 Overview of the TabPFN Benchmark Evaluation Pipeline and Data Sources. (**a**.) Example excerpt from the FOSP dataset, showing patient-level clinical data including diagnosis date, ICD-O-3 code, morphology, sex, and survival outcome. Consistent structured data formats were used across all datasets to establish the binary classification tasks. (**b**.) Patient-level tabular data were collected from three primary sources: the FOSP dataset, the SEER registry, and datasets published by Arasteh et al. These included both cancer-related and non-cancer conditions. Each dataset defines a binary classification task, with target labels representing outcomes such as disease presence or survival at specific time points. (**c**.) All datasets underwent a standardized preprocessing pipeline, including missing value imputation, and one-hot encoding. A comprehensive suite of machine learning models including boosting models, classic algorithms, and TabPFN with and without HPO was trained and evaluated on the processed data. Model performance was assessed over 100 bootstrap iterations using metrics such as AUROC, 95% confidence intervals, and runtime per model iteration

settings while minimizing confounding effects from dataset-specific evaluation procedures.

Comparative discrimination performance across clinical tasks

Across the twelve clinical prediction tasks, TabPFN achieved discrimination performance comparable to strong ML baselines, with small, task-dependent differences (Fig. 2). When comparing AUROC values for TabPFN, the best-performing ML model, and reported literature benchmarks, all three were closely aligned across tasks (Fig. 2a). For each given task, the best-performing ML model is defined as the model that had the highest AUROC for that task, as listed in Table 1 and annotated on Fig. 2a.

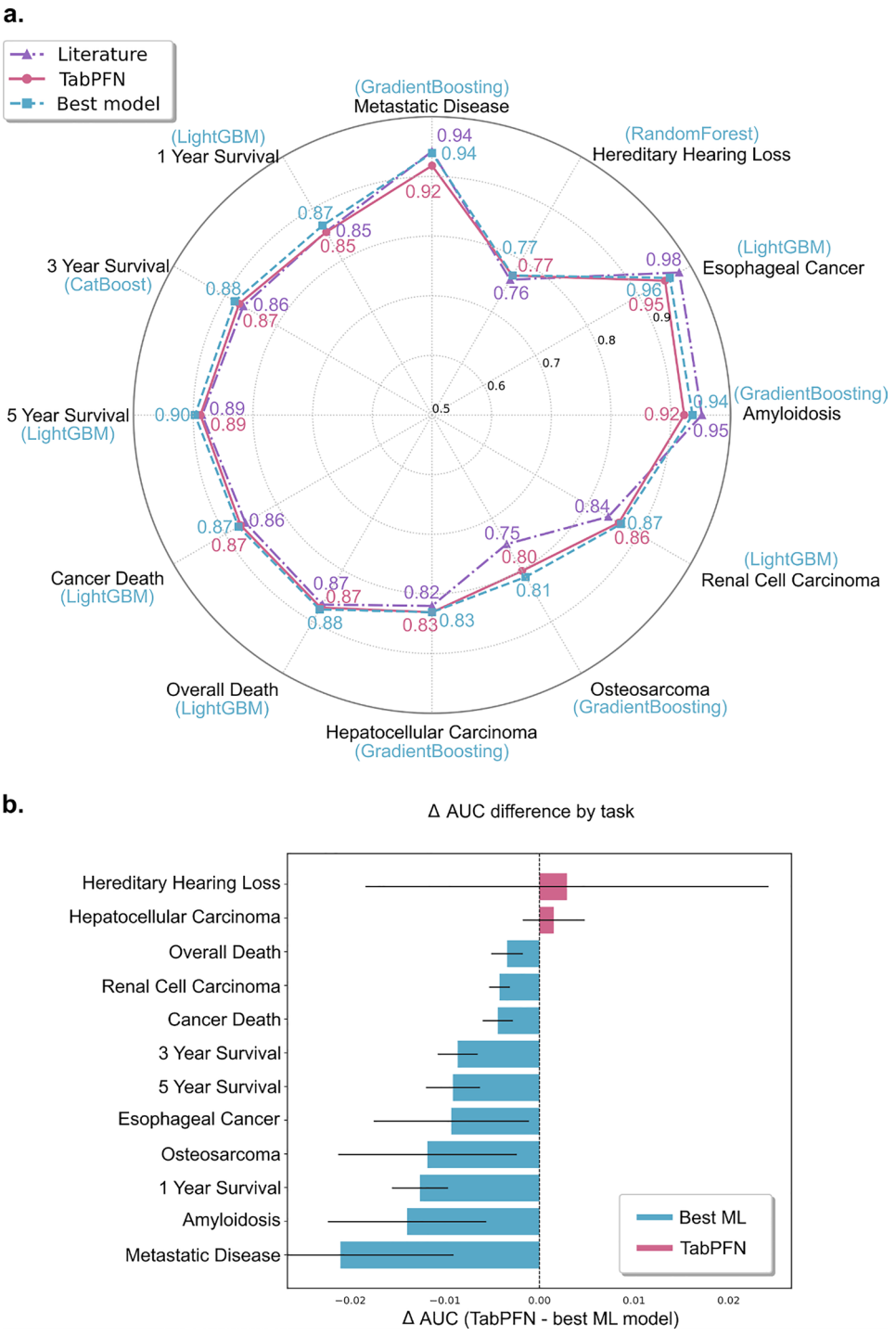


Fig. 2 Radar Plot of comparative TabPFN, Machine Learning, and Literature AUROC performance. (**a**.) AUROC across multiple clinical prediction tasks for three benchmarks: TabPFN (pink solid line), the best-performing machine learning model from the benchmark suite (blue dashed line), and reported values from the literature (purple dash-dot line). Prediction tasks span cancer and non-cancer domains, including survival at 1, 3, and 5 years, cancer-specific and overall mortality, as well as disease classification tasks such as Amyloidosis and Hereditary Hearing Loss. TabPFN generally matched the strongest baseline, with small task-dependent differences. (**b**.) Error bars indicate the 95% CI for ΔAUROC derived from the corresponding 95% AUROC CIs. Positive values (pink) indicate superior performance by TabPFN, while negative values (blue) indicate cases where the ML model performs better. In Fig. 2b, ΔAUROC intervals for Best ML–favored tasks do not cross zero, consistent with robust ML advantages for those tasks

Table 1 Best-performing ML models per target

Target	Best-performing Model
Metastatic Disease	GradientBoosting
Hereditary Hearing Loss	RandomForest
Esophageal Cancer	LightGBM
Amyloidosis	GradientBoosting
Renal Cell Carcinoma	LightGBM
Osteosarcoma	GradientBoosting
Hepatocellular Carcinoma	GradientBoosting
Overall Death	LightGBM
Cancer Death	LightGBM
5 Year Survival	LightGBM
3 Year Survival	CatBoost
1 Year Survival	LightGBM

Best-performing ML model for each clinical target, with LightGBM emerging most frequently, followed by GradientBoosting, while RandomForest and CatBoost performed best for select outcomes

In direct comparisons against the strongest baseline model per task, TabPFN outperformed the best ML model in 2 of 12 tasks and underperformed in the remaining 10 tasks when considering 95% confidence intervals, with no cases of overlapping intervals. However, the magnitude of these differences was small. Across tasks, most absolute differences in AUROC between TabPFN and the best ML model were within ± 0.02 (Fig. 2b), indicating near-parity in discrimination performance.

Both TabPFN and the strongest baseline models, most frequently boosting or bagging approaches such as CatBoost and ExtraTrees, achieved AUROC values close to those reported in the original publications, with only minor deviations observed in isolated cases, including the SEER osteosarcoma task (Fig. 2a). Reported literature AUROCs are included for reference and may not be directly comparable across differing evaluation protocols. Negative-control label permutations yielded AUROC values near 0.5 for all models (Supplementary Fig. S1), indicating that the observed results were unlikely to reflect a false signal. Per-model performance summaries are provided in the Supplement (Bagging: Supplementary Fig. S2; Boosting: Supplementary Fig. S3; Classic: Supplementary Fig. S4; and TabPFN-HPO and MLP: Supplementary Fig. S5).

Performance-efficiency trade-offs

We next examined computational efficiency in relation to discrimination performance (Fig. 3). When AUROC and runtime were normalized to logistic regression as a reference, clear differences emerged across model families in the performance-efficiency plane.

Despite achieving AUROC values comparable to the strongest ML baselines, TabPFN exhibited substantially higher computational cost. Across tasks, TabPFN had a median runtime 5.53-fold higher than the best-performing ML model. In contrast, boosting and bagging methods most frequently occupied a favorable region characterized by similar AUROCs with lower runtimes.

Classic models and multilayer perceptrons generally showed less favorable trade-offs, either due to lower discrimination performance or higher runtime. Across the benchmark, TabPFN did not demonstrate a consistent accuracy advantage sufficient to offset its higher computational cost, placing it in a less favorable region of the performance-efficiency space.

Given TabPFN's near-parity discrimination with strong ML baselines, we assessed whether comparable performance reflected similar reliance on input features. Permutation SHAP on cardiac amyloidosis prediction revealed a steep long-tailed importance profile for TabPFN, with most attribution mass concentrated in a small subset of predictors (Fig. 4a). Comparing SHAP-based rankings between TabPFN and each baseline showed moderate-to-high agreement, with rank-biased overlap (RBO) values around ~ 0.55–0.65 and consistent overlap among the top-20 features (typically ~ 7–10 shared features; Fig. 4b). Overall predictive performance was similar, TabPFN generally emphasized the same top predictors as boosting models, with higher computational costs.

Discussion

We conducted a systematic benchmark of TabPFN against established ML models across a diverse set of clinically relevant binary prediction tasks, evaluating both its default configuration and its automated HPO. Across tasks, TabPFN achieved discrimination performance comparable to strong ML baselines but did not consistently improve predictive accuracy. In contrast, TabPFN introduced a clear computational trade-off, with substantially longer runtimes than competing approaches. While CPU-only execution was feasible, runtimes were markedly prolonged, whereas bagging and boosting methods frequently achieved similar AUROC values with substantially lower computational cost and without reliance on specialized hardware.

These findings are consistent with prior work showing that ensemble-based methods remain strong baselines for binary prediction in clinical settings [35]. In addition to competitive performance, boosting and bagging models benefit from mature tooling for feature attribution and error analysis, which facilitates model review and clinical validation [36]. Their relative simplicity also supports deployment on standard CPU-based infrastructure, which remains a practical consideration in many clinical environments.

The absence of a consistent performance advantage for TabPFN may reflect differences between the data regimes

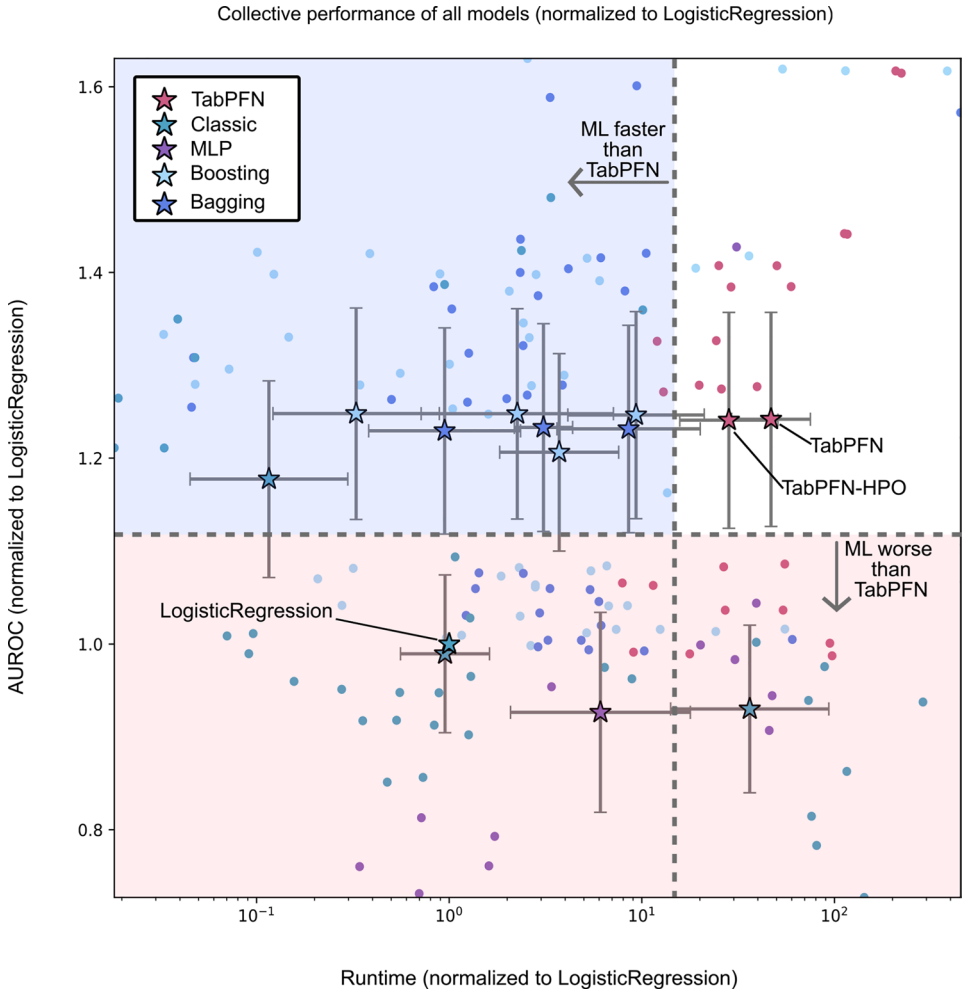


Fig. 3 Runtime vs. AUROC normalized to LogisticRegression. Model performance by runtime (x-axis, log scale) and AUROC (y-axis), both normalized to LogisticRegression (reference at [1,1], bolded). Stars represent mean performance for each model class (excluding LogisticRegression reference); TabPFN, boosting, bagging, classic, and multilayer perceptron, all with 95% confidence intervals shown as error bars. Circles denote individual predictions across clinical targets contributing to class means

used for model development and those encountered in real-world clinical applications. TabPFN was trained primarily on large collections of synthetic datasets, which may not fully capture the heterogeneity, noise structure, and distributional shifts present in clinical tabular data. This interpretation aligns with broader observations in healthcare ML cautioning against replacing well-tuned classical models solely on the basis of architectural novelty [37].

The observed performance patterns may also reflect practical constraints related to TabPFN's current column/row processing design, which limited the effective use of very large cohorts in this study and may have influenced achievable performance. In addition, the present benchmark focused exclusively on binary classification tasks and did not evaluate multi-class or regression settings, despite TabPFN's broader capabilities. The FOSP Overall Death and Cancer Death endpoints were also reproduced as binary endpoint tasks, without modeling follow-up time or censoring, and should therefore not be interpreted as survival-modeling benchmarks. Future benchmarking efforts that extend to larger tables, additional task formulations, and non-clinical domains will be necessary to more fully assess the potential of tabular foundation models.

Taken together our findings suggest that for large scale life science datasets, and the binary clinical classification tasks we evaluated here, TabPFN should be considered a complementary method rather than a clear replacement for established ML models, especially when computational efficiency and deployment constraints are critical. In clinical workflows, scalability, interpretability, and operational simplicity are often as important as marginal differences in discrimination performance. Compared with widely used gradient-boosting frameworks such as CatBoost or LightGBM, TabPFN currently offers

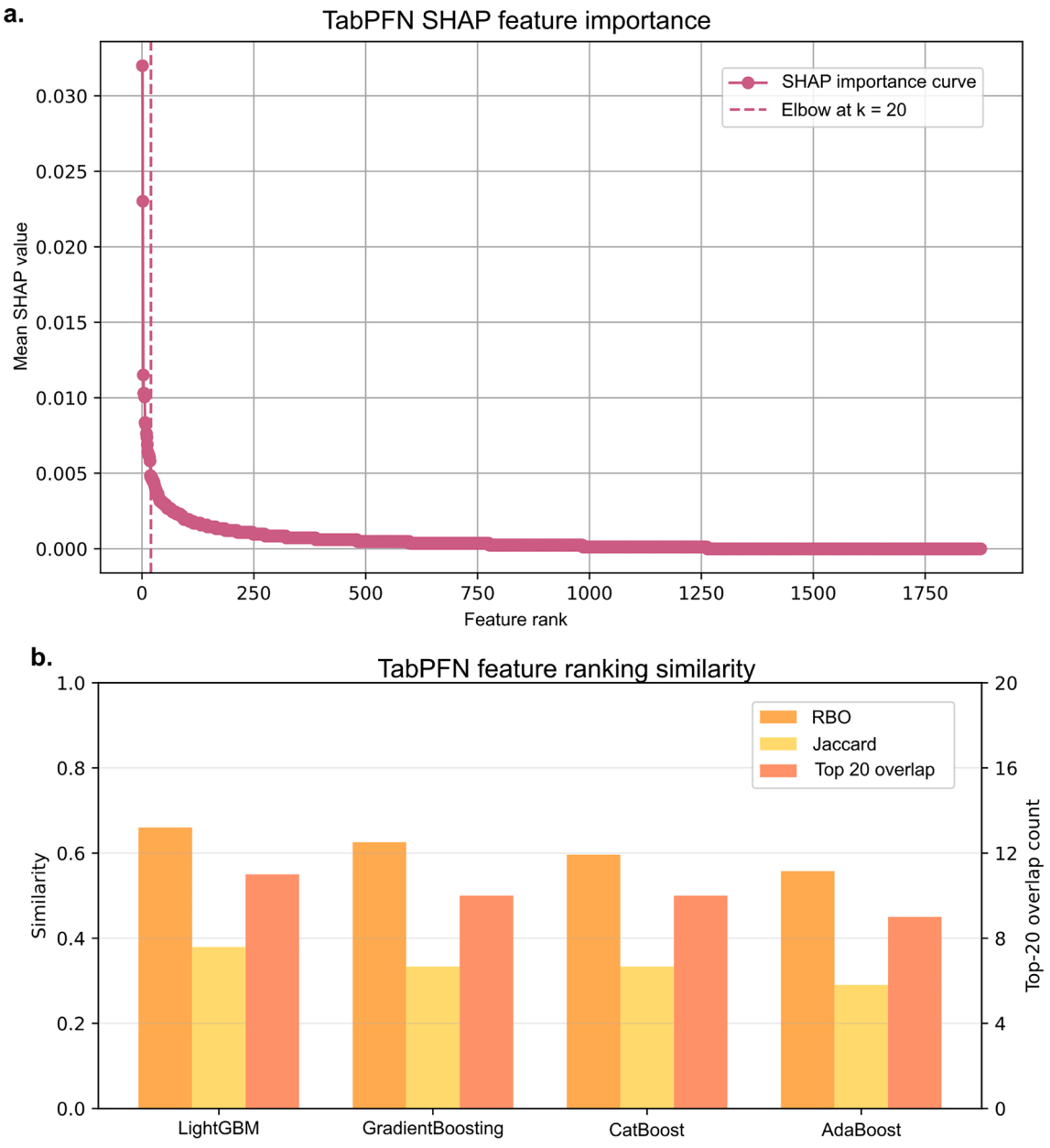


Fig. 4 Explainability agreement between TabPFN and classical baselines. (**a**.) Global TabPFN feature-importance curve computed from permutation SHAP. Points show features sorted by decreasing mean absolute SHAP value; the dashed vertical line marks the cutoff k = 20 illustrating the rapid drop-off in importance beyond the highest-ranked features. (**b**.) Similarity between TabPFN's SHAP-based feature ranking and each baseline model's ranking predicting cardiac amyloidosis. Rank overlap quantified by rank-biased overlap (RBO) (left y-axis indicates similarity among top-ranked features), set overlap by Jaccard similarity on the top-20 features, and absolute agreement by the top-20 overlap count (right y-axis; maximum 20)

limited flexibility for domain-specific objectives and constraints. While post-hoc explainability methods can be applied to any model, tree-based ensembles often remain easier to audit in practice due to well-established workflows for attribution and error analysis. In our analysis, TabPFN and Boosting ML models produced similar SHAP-derived importance rankings, implying substantial overlap in the signals each model exploited. Taken together, these results suggest that TabPFN does not consistently offer added benefit for feature discovery or model auditing relative to baselines, at least under post-hoc SHAP attribution analyses.

For foundation models to achieve meaningful adoption in healthcare, they must demonstrate clear and consistent advantages that extend beyond architectural novelty. Although TabPFN may be useful as an out-of-the-box baseline with minimal feature engineering or tuning, broader clinical adoption will depend on improvements in computational efficiency, scalability, and generalizability across diverse clinical tasks and data environments.

Methods

Datasets

We constructed twelve clinically relevant binary classification tasks using eight publicly available datasets with established use in the literature. From the Fundação Oncocentro de São Paulo (FOSP), previously used by Woźniacki et al. [38], we derived five survival-related targets: 1-year survival, 3-year survival, 5-year survival, cancer-related deaths, and overall deaths. FOSP data were acquired from the publicly accessible Fundação Oncocentro Registry [39], and colorectal cancer cases were identified using ICD-O-3 topography codes C18–C20. FOSP Overall Death and Cancer Death were analyzed as reproduced binary endpoint tasks mirroring previous publications, without modeling follow-up time or censoring.

Arasteh et al. provided four tabular clinical prediction datasets for cardiac amyloidosis, esophageal cancer, hereditary hearing loss, and metastatic disease. These datasets were downloaded from the supplementary materials of the original benchmark study and used as provided, including predefined splits and engineered features. The tasks corresponded to prediction of amyloidosis risk [40], hereditary hearing loss inheritance [41], high-risk esophageal cancer [42], and cancer metastasis [43].

From the Surveillance, Epidemiology, and End Results (SEER) program [44], we reproduced three previously published ML tasks: hepatocellular carcinoma (HCC) metastasis [45], osteosarcoma (OST) [46], and renal cell carcinoma (RCC) survival [47]. SEER data were extracted using SEER*Stat (v8.4.5) by replicating published inclusion criteria: (HCC metastasis; "Incidence—SEER Research Data, 17 Registries, Nov 2023 Submission, 2000–2021"; Primary Site C22), (OST survival; "Incidence—SEER Research Data, 17 Registries, Nov 2024 Submission, 2000–2022"; Primary Site C40.0–C41.9), (RCC 5-year overall survival; same 2024 submission; Site recode ICD-O-3/WHO 2008 = Kidney and Renal Pelvis).

All datasets were structured for clinically relevant binary classification tasks.

Experimental design

For benchmark comparison, twelve supervised classifiers implemented in *scikit-learn* were evaluated using a fixed random seed for reproducibility. These models included AdaBoost, Bagging, CatBoost, DecisionTree, ExtraTrees, GradientBoosting, K-Nearest Neighbors (KNN), LightGBM, LogisticRegression, RandomForest, Multilayer Perceptron (MLP), and Support Vector Classification (SVC).

All experiments were implemented in Python 3.12.7 using TabPFN v2.5 and *tabpfn-extensions* v0.1.2, which supports automated HPO configuration. For each prediction task, TabPFN's HPO function was allowed 50 iterations to autoselect optimal settings. All models were trained and evaluated on the same features, outcome definitions, and tasks-specific protocols to ensure fair comparison of predictive performance and runtime across models and tasks.

Each clinical target had previously reported AUROC values in the literature, which were reproduced to serve as positive controls. In addition to the default TabPFN configuration, we evaluated the recently introduced TabPFN HPO feature released after the original TabPFN publication (https://github.com/PriorLabs/tabpfn-extensions).

General preprocessing

Across all datasets, minimal preprocessing was applied to maintain comparability. Standard preprocessing steps included missing value handling, categorical encoding, date transformation, outcome engineering, and exclusion of target leakage features.

Missing values in numerical columns were imputed using a constant "-9" and cast to a floating-point format to keep consistent with HCUP recommendations [48]. Missing datetime entries were filled with the placeholder date "1900-01-01", and missing categorical values labeled as "Missing". Columns with more than 95% missing values or with (near-)constant (i.e., had ≤ 1 unique non-null value) were removed. High cardinality categorical columns (> 100 unique values) were also excluded to reduce unnecessary model complexity and hinder generalization.

Columns containing date-like strings (YYYY-MM-DD format) were automatically detected and converted to numeric representations as days from 1900-01-01. Target variables and derived outcome-related features, such as "days_survived", were excluded from all preprocessing steps prior to model training to prevent data leakage. A full list of excluded features is provided in the processing scripts (Supplemental List 1).

Dataset-specific preprocessing

FOSP data were filtered to colorectal cancer patients using TOPO codes C18–C20. Outcome variables were defined as follows: overall_death = 1 if ULTINFO = 3 or 4, and cancer_death = 1 if ULTINFO = 3. Survival targets (1-year, 3-year, and 5-year) were computed from DTDIAG and DTULTINFO.

The four datasets from Arasteh et al. (amyloidosis, esophageal cancer, hereditary hearing loss, metastatic

disease) were used as provided, including predefined splits and engineered features.

SEER datasets were extracted using disease-specific criteria as described in the Datasets section. HCC data came from the 2023 submission (2000–2021) and were filtered by Primary Site = C22. For OST cases, we used the 2024 submission (2000–2022) with Primary Site = C40.0–C41.9. RCC cases were drawn from the same 2024 submission, filtered using Site recode ICD-O-3/WHO 2008 = Kidney and Renal Pelvis.

Sampling and bootstrapping

For each of the twelve clinical prediction tasks, data were preprocessed and features (X) and labels (y) were separated. After one-hot encoding, all datasets remained within the input constraints of TabPFN v2.5, and no dimensionality reduction was required. Data were split into training and testing sets using a 70/30 split. In the same fashion, any datasets with more than 100,000 samples were randomly sampled down to 100,000 samples to allow for use within TabPFN 2.5's input constraints.

Model evaluation followed a nested-loop design. In the outer loop, each baseline classifier was evaluated, while the inner loop applied 100 bootstrap resamplings to the training data for each model–target pair. The test set remained fixed. After 100 train/test cycles per model and task, an additional 100 iterations were run as negative controls. We used a fixed stratified split and bootstrapped the training set only; uncertainty therefore reflects sampling variability conditional on the held-out test set. Hyperparameters were selected once per task using 50 random search hyper parameter setting iterations and then fixed across bootstrap resamplings. Hyperparameter setting range for all baseline ML models are listed in Table 2.

A second evaluation loop, identical in structure, replaced the baseline models with TabPFN and TabPFN-HPO. Hyperparameters were selected once per task using 50 optimization iterations and then fixed across bootstrap resamplings. All models were evaluated using identical features, targets, and preprocessing. Baseline models were run on CPU, whereas TabPFN models were evaluated using an NVIDIA A100 40GB GPU. AUROC and runtime metrics were collected for all runs.

Evaluation

Performance metrics included AUROC and runtime, computed per model and per prediction task across 100 bootstrap iterations. Runtimes were measured using system time at the start and end of each loop, then averaged across runs for each model-target pair.

Mean AUROCs with 95% confidence intervals were computed, and TabPFN was compared against both the best-performing baseline and reported literature AUROC on a per-task basis. To assess performance–efficiency trade-offs, AUROC and inference time were normalized to LogisticRegression, enabling direct comparison of model quality versus runtime. No class rebalancing or oversampling was applied, preserving the real-world label imbalance of each dataset.

All threshold-based metrics including balanced accuracy and F1-score, were computed after re-thresholding using the Youden index. Additional metrics, including AUPRC and standard deviations, are reported in Supplemental Table 2.

To compare explainability across models, we computed permutation SHAP for TabPFN and Boosting ML baselines on predicting cardiac amyloidosis using the model's predicted class as the output. SHAP was estimated on a random sample of up to 20 test instances with a background reference set of up to 1877 features. Global feature importance was summarized as mean absolute SHAP per feature and used to derive ranked feature lists. We quantified agreement between TabPFN and each baseline using RBO, Jaccard similarity on the top 20 features, and the absolute overlap count. To compare GPU vs. CPU dependent run times, AdaBoost and CatBoost were both run on CPU and GPU with otherwise identical settings (Supplemental Fig. 6).

Table 2 Machine learning models and hyperparameter settings

Model	Hyperparameter Search Space
AdaBoost	n_estimators: [25, 50, 75, 100, 150, 200] learning_rate: [0.01, 0.05, 0.1, 0.5, 1.0, 1.5]
Bagging	n_estimators: [25, 50, 75, 100, 150, 200] max_samples: [0.5, 0.7, 0.85, 1.0] max_features: [0.5, 0.75, 1.0]
CatBoost	iterations: [50, 100, 200, 300, 500] learning_rate: [0.01, 0.03, 0.05, 0.1, 0.2] depth: [4, 6, 8, 10] l2_leaf_reg: [1, 3, 5, 7, 9]
Decision Tree	max_depth: [None, 2, 3, 4, 5, 6, 8, 10] min_samples_split: [2, 5, 10, 20] min_samples_leaf: [1, 2, 4, 8]
Extra Trees	n_estimators: [50, 100, 150, 200, 300] max_depth: [None, 3, 5, 8, 12] min_samples_split: [2, 5, 10] min_samples_leaf: [1, 2, 4] max_features: [sqrt, log2, None, 0.5]
Gradient Boosting	n_estimators: [50, 100, 150, 200, 300] learning_rate: [0.01, 0.03, 0.05, 0.1, 0.2] max_depth: [1, 2, 3, 4, 5] subsample: [0.6, 0.8, 1.0]
KNN	n_neighbors: [3, 5, 7, 9, 11, 15, 21] weights: ["uniform", "distance"] p: [1, 2]
LightGBM	n_estimators: [50, 100, 150, 200, 300, 500] learning_rate: [0.01, 0.03, 0.05, 0.1, 0.2] num_leaves: [15, 31, 63, 127] max_depth: [-1, 3, 5, 7, 10] min_child_samples: [5, 10, 20, 50] subsample: [0.6, 0.8, 1.0] colsample_bytree: [0.6, 0.8, 1.0]
LogisticRegression	max_iter: [100, 200, 300, 500, 700, 1000] C: [0.01, 0.1, 0.5, 1.0, 2.0, 5.0, 10.0]
RandomForest	n_estimators: [25, 50, 100, 150, 200] max_depth: [2, 3, 4, 5, 6, 8, 10, None] min_samples_split: [2, 5, 10, 20] min_samples_leaf: [1, 2, 4, 8] max_features: ["sqrt", "log2", None, 0.5]
MLP	hidden_layer_sizes: [(50,), (100,), (150,), (100, 50), (200,), (200, 100)] max_iter: [200, 500, 800, 1000] activation: ["relu", "tanh"] alpha: [1e-5, 1e-4, 1e-3, 1e-2] learning_rate_init: [1e-4, 5e-4, 1e-3, 5e-3, 1e-2]
SVC	C: [0.1, 0.5, 1.0, 2.0, 5.0, 10.0] gamma: ["scale", "auto", 0.001, 0.01, 0.1, 1.0]

This table lists the machine learning algorithms in order of use in the study along with their corresponding hyperparameter configuration ranges used in random search

Abbreviations

AI Artificial intelligence
AUROC area under the receiver operating characteristic
EHR electronic health record
FM foundation model
FOSP Fundação Oncocentro de São Paulo
HCC hepatocellular carcinoma
HPO hyper-parameter optimization
JSC JUWELS at Jülich Supercomputing Centre
KNN K-Nearest Neighbors
ML machine learning
MLP Multilayer Perceptron
NIC Neumann Institute for Computing
OST osteosarcoma
RBO rank-biased overlap
RCC renal cell carcinoma
SEER Surveillance, Epidemiology, and End Results
SVC Support Vector Classification
TabPFN Tabular Prior-data Fitted Network

Supplementary Information

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Supplementary Material 1

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Author contributions
L.A.S., M.G., L.H., and J.N.K. contributed to conceptualization. L.A.S. and J.N.K. contributed to Data Curation. L.A.S. conducted formal analysis. J.N.K. conducted funding acquisition. L.A.S., M.G., L.H., J.L., and J.N.K. contributed methodology. L.A.S. and J.N.K. contributed to project administration and supervision. L.A.S., M.G., J.N.K. contributed to visualization. L.A.S., M.G., T.L., J.L., Z.I.C., and J.N.K. contributed to writing the manuscript.

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Data availability
No new datasets were generated for this study. All datasets analyzed are publicly available from the original repositories/databases: Fundação Oncocentro de São Paulo (FOSP;https://fosp.saude.sp.gov.br/), the public datasets curated by Arasteh et al. for cardiac amyloidosis, esophageal cancer, hereditary hearing loss, and metastatic disease (https://www.nature.com/articles/s41467-024-45879-8), and the Surveillance, Epidemiology, and End Results (SEER) Program (https://seer.cancer.gov/). Dataset-specific links and identifiers are provided in Supplementary Table 1, including the amyloidosis source article, the esophageal cancer GitHub repository, the hereditary hearing loss Mendeley Data record, the metastatic disease Zenodo record, and the FOSP and SEER registry links. SEER-derived cohorts were extracted using SEER*Stat v8.4.5 from the Nov 2023 and Nov 2024 SEER Research Data submissions as described in Methods. All analysis code is available at https://github.com/KatherLab/tabpfnbenchmark under an MIT licence. The code was developed specifically for this study.

Declarations

Ethical consideration
All datasets used in this study were publicly available and fully de-identified prior to access. No new patient data were collected, and no identifiable information was handled at any point. Because all sources are open-access and de-identified, this work did not require institutional ethics approval or informed consent. All data analyses were performed in accordance with the data-use terms of the respective repositories and the principles of the Declaration of Helsinki.

Consent for publication
Not applicable.

Competing interests
J.N.K. declares ongoing consulting services for AstraZeneca and Bioptimus. Furthermore, he holds shares in StratifAI, Synagen, and Spira Labs, has received an institutional research grant from GSK and AstraZeneca, as well as honoraria from AstraZeneca, Bayer, Daiichi Sankyo, Eisai, Janssen, Merck, MSD, BMS, Roche, Pfizer, and Fresenius. M.G. has received honoraria for lectures sponsored by Sartorius AG and AstraZeneca. All other authors do not have anything to disclose.

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